

Customer Behavior Analysis Project Report

Executive Summary

1. Project Overview & Objectives

This project aims to analyze the shopping behavior data of e-commerce customers (3.9K customers) to discover purchasing patterns, evaluate customer loyalty, and develop data-driven marketing strategies to increase revenue.

2. Workflow & Tools Used

- **Data Processing & Cleaning (Python - Pandas):** Handling missing values using the median, standardizing column names to facilitate analysis, feature engineering by segmenting ages into 4 main categories, and removing duplicate columns.
- **Database Management (MySQL):** Creating tables and uploading clean data to execute complex queries and extract metrics and figures.
- **Data Visualization (Power BI):** Designing a professional interactive dashboard to clearly display insights and sales targets for decision-makers.

3. Key Analyses & Answered Questions (SQL)

- **Revenue Contribution by Gender:** Determining purchasing power and sales distribution between females and males for targeted marketing.
- **Subscription Program Evaluation:** Comparing average spending and total revenue between subscribers and non-subscribers to evaluate return on investment.
- **Customer Segmentation:** Classifying customers into categories (new, returning, loyal) based on purchase history to personalize offers.

- **Demographic Analysis of Age Groups:** Studying and evaluating the financial contribution of each age group (young adults, adults, middle-aged, seniors).

4. Interactive Dashboard & Outputs

✓ **Total Customers:** 3.9K | **Average Purchase Amount:** \$59.76 | **Average Rating:** 3.75

✓ **Top Revenue Category:** The Clothing category achieved the highest revenue of \$104K.

✓ **Top Spending Age Group:** Young Adult category with a total revenue of \$62K.

